

ROSALIND AGCM, AB, Canada

WMO: 712020

Lat: 52.7923N

Lon: 112.4228W

Elev: 708

StdP: 93.10

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-30.9	-27.9	-34.3	0.2	-30.8	-31.3	0.2	-28.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	28.7	16.7	26.9	16.4	25.3	15.8	19.1	25.1	18.0	24.5	17.0	23.2	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	13.0	22.3	15.5	12.0	20.8	14.5	11.3	19.7	57.3	25.0	53.6	24.6	50.4	23.3	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-34.3	32.0	3.0	2.4	-36.5	33.7	-38.3	35.1	-40.0	36.4	-42.2	38.1		
(5) N/A	N/A	N/A	-34.4	21.7	3.0	1.6	-36.6	22.9	-38.4	23.8	-40.1	24.6	-42.2	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-10.1	-11.2	-4.7	2.9	10.7	14.6	16.8	15.9	11.2	4.0	-4.6	-11.0		
	DBStd	11.97	8.15	8.38	7.51	5.29	4.40	2.93	2.59	3.03	4.55	4.91	7.15	7.73		
	HDD10.0	3263	622	594	456	217	46	3	0	1	40	194	439	652		
	HDD18.3	5645	880	827	714	462	238	116	62	87	216	444	688	911		
	CDD10.0	691	0	0	0	5	67	141	211	183	76	8	0	0		
	CDD18.3	31	0	0	0	0	1	4	14	9	2	0	0	0		
	CDH23.3	842	0	0	0	3	82	100	263	263	129	2	0	0		
	CDH26.7	176	0	0	0	0	13	12	56	62	33	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	423	21	12	18	31	47	79	77	57	34	19	15	14		
	PrecMax	529	63	26	38	60	85	114	166	107	77	45	56	30		
	PrecMin	289	1	3	2	4	16	25	24	6	3	5	3	5		
	PrecStd	71	15	7	9	16	21	26	34	24	20	9	12	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.5	6.2	14.0	22.4	28.4	28.0	30.6	30.4	30.4	22.8	14.7	6.3		
		MCWB	2.7	3.3	7.3	10.2	13.5	16.5	19.6	16.8	15.7	12.3	7.5	2.9		
	2%	DB	4.5	3.9	9.8	18.7	25.6	26.1	28.2	28.4	27.1	18.2	9.7	4.1		
		MCWB	1.8	1.3	4.7	8.7	12.6	15.9	18.4	16.9	14.5	10.3	4.9	1.4		
	5%	DB	3.1	2.3	6.5	15.2	23.2	24.1	26.3	26.3	24.1	15.2	6.5	2.0		
		MCWB	1.1	0.2	3.1	6.8	11.8	15.1	17.9	16.8	13.9	8.4	2.9	-0.2		
	10%	DB	1.4	0.5	4.1	12.2	20.4	22.1	24.3	24.3	21.0	12.4	4.2	-0.5		
		MCWB	-0.4	-1.0	1.5	5.6	10.7	14.1	16.9	16.1	12.4	7.0	1.6	-2.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.3	7.6	11.0	14.6	18.7	21.5	20.2	16.7	12.9	8.3	3.2		
		MCDB	5.8	6.1	12.7	22.1	24.9	25.1	27.1	25.2	26.2	21.1	13.9	6.1		
	2%	WB	2.1	1.6	4.9	8.9	13.3	17.1	19.6	18.6	15.3	10.9	5.3	1.5		
		MCDB	4.3	3.6	9.2	17.9	22.9	23.5	25.6	24.9	24.8	16.7	9.0	3.9		
	5%	WB	1.1	0.4	3.0	7.3	12.4	15.9	18.5	17.5	14.1	9.0	3.3	-0.2		
		MCDB	3.1	2.1	6.0	14.3	21.4	22.0	24.8	24.4	22.4	14.3	6.0	1.8		
	10%	WB	-0.4	-1.0	1.6	5.8	11.4	14.8	17.6	16.5	12.9	7.4	1.8	-2.3		
		MCDB	1.4	0.5	3.9	11.5	19.4	20.6	23.6	23.2	20.2	12.0	3.7	-0.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	10.4	10.3	12.1	13.9	12.3	13.1	14.0	14.3	12.2	9.7	8.9		
		MCDBR	10.1	10.9	12.8	18.3	18.4	15.6	16.5	18.4	20.6	17.2	12.7	10.2		
		MCWBR	8.2	8.8	8.5	9.5	8.1	7.0	8.5	8.8	9.5	9.6	8.4	8.1		
		MCDBR	9.9	10.4	12.2	16.8	16.2	14.1	15.2	16.0	18.9	15.8	11.5	10.0		
	5% WB	MCWBR	8.1	8.5	8.4	9.2	7.5	7.2	8.7	8.5	9.1	9.2	7.9	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.271	0.297	0.330	0.349	0.362	0.358	0.370	0.314	0.283	0.255	0.247		
	taud		2.280	2.358	2.392	2.363	2.365	2.352	2.396	2.340	2.466	2.530	2.454	2.302		
	Ebn at Noon		730	855	900	905	901	888	886	852	871	838	761	689		
	Edh at Noon		61	77	92	108	114	117	110	110	85	65	51	49		
(44) All-Sky Solar Radiation	RadAvg		1.05	2.10	3.55	4.71	5.75	6.06	6.30	5.16	3.58	2.12	1.06	0.77		
	RadStd		0.10	0.17	0.26	0.45	0.38	0.39	0.32	0.36	0.38	0.25	0.11	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (89 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page